

PROBLEM SET 4 : THE REAL EXCHANGE RATE AND PURCHASING POWER PARITY

CHAPTER 9 IN SCHMITT-GROHÉ, URIBE AND WOODFORD

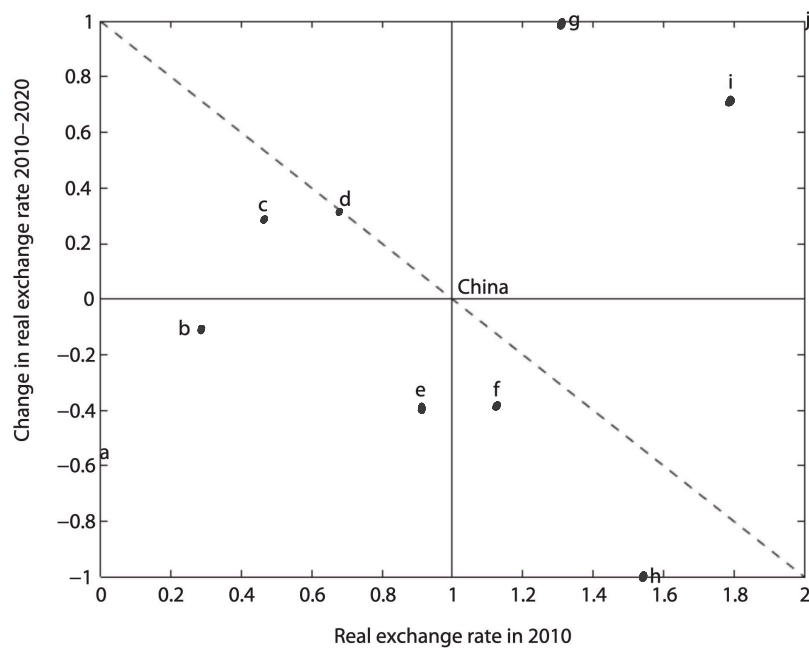
Version 1.0 - 10/01/2026

True/False/Uncertain

1. The law of one price holds for football game tickets.
2. Rich countries are more expensive than poor countries.
3. Home bias means that productivity is different between countries

Exercise 9.1 (International Price Convergence)

The figure below displays the real exchange rate of 10 hypothetical countries relative to China in 2010 and the change in the real exchange rate between 2010 and 2020.



1. What is the meaning of the broken line?
2. What countries were less expensive than China in 2010 and became even cheaper after 10 years?

3. What countries, if any, were more expensive than China in 2010, but less expensive than China in 2020?
4. What countries, if any, were less expensive than China in 2010, but more expensive than China in 2020?
5. What countries, if any, became as expensive as China in 2020?

Exercise 9.5 (Nontradable Goods and PPP)

In countries A and B, households consume only wheat and haircuts. Wheat is internationally traded, whereas haircuts are not. In country A, a ton of wheat is 10 times as expensive as a haircut, while in country B it is only 5 times as expensive. The consumer price index in country $i = A, B$, denoted P_i , is $P_i = (P_i^W)^{0.25}(P_i^H)^{0.75}$, where P_i^W and P_i^H denote the prices of wheat and haircuts in country i

1. Calculate the real exchange rate between countries A and B.
2. Which country is more expensive and why?

Exercise 9.13 (Microfounded Price Indices)

Let C , C_T , and C_N denote consumption, consumption of tradables, and consumption of nontradables. Let P , P_T , and P_N denote the consumer price level, the price of tradables, and the price of nontradables. Find the consumer price level under the following aggregator functions

1. Cobb-Douglas aggregator:

$$C = C_T^\gamma C_N^{1-\gamma} \text{ with } 0 < \gamma < 1.$$

2. Leontief aggregator:

$$C = \min \left\{ \frac{C_T}{\gamma} \frac{C_N}{1-\gamma} \right\} \text{ with } 0 < \gamma < 1.$$

3. Linear aggregator:

$$C = \gamma C_T + (1-\gamma)C_N \text{ with } 0 < \gamma < 1.$$